

A Thermally Tunable, Magnetically Actuated Soft Haptic Unit Cell

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RESEARCH QUESTION

Can thermal conditioning enable localized deformation under remote magnetic actuation?

MOTIVATION

Magnetic-silicone fabrication [1] and magneto-origami [2] establish routes to soft magnetic actuation. Thermomagnetic PCL/iron systems demonstrate thermal reconfiguration and shape retention [3]. Pre-buckled ferroelastomer beams demonstrate magnetically triggered bistable switching [4]. Here, a haptic unit cell targets retention without continuous holding power.

01 Unit-cell architecture

Polycaprolactone (PCL) surrounds a magnetized NdFeB/Ecoflex composite.

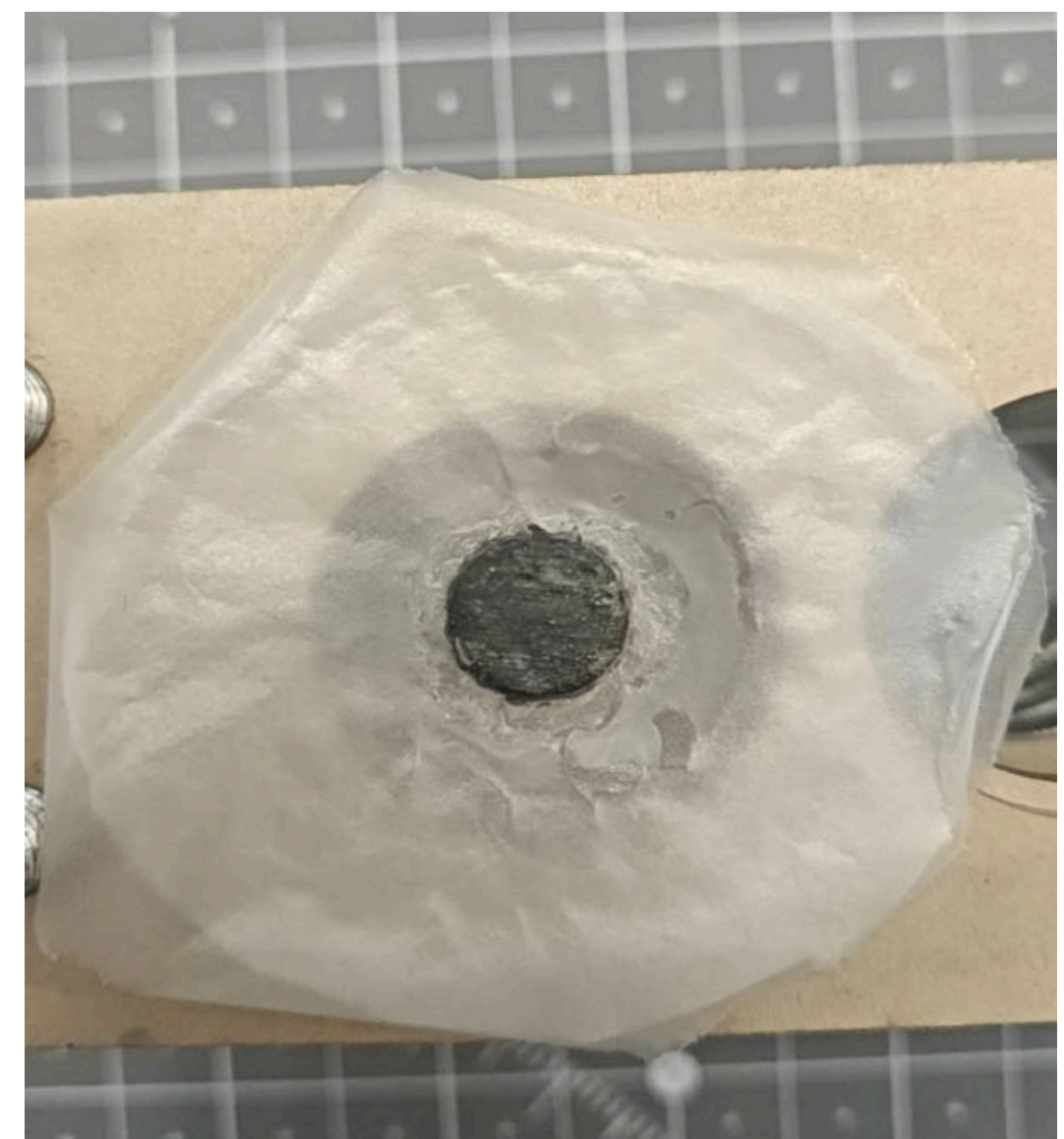
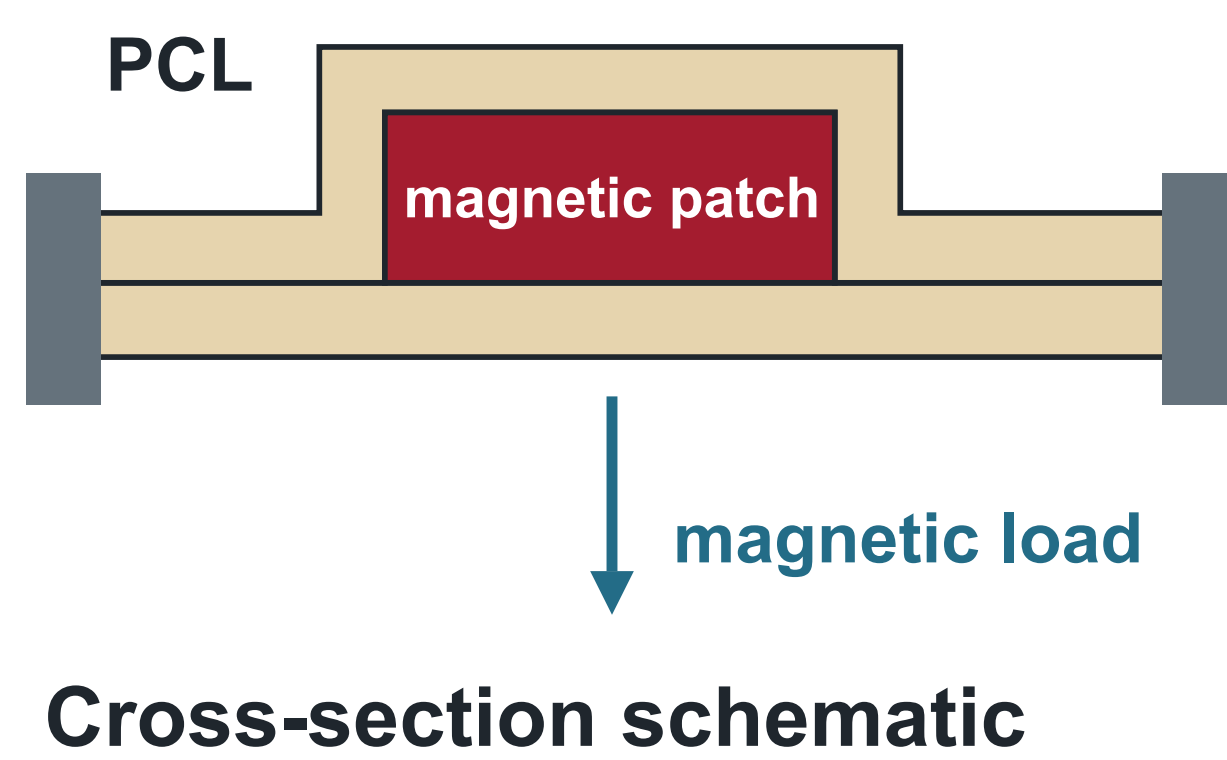


Figure 1. Prototype photograph. Physical unit cell showing the central magnetic patch within the surrounding PCL structure.



Cross-section schematic

Figure 2. Unit-cell architecture. Schematic cross-section of the PCL structure, magnetic patch and edge support. Not to scale.

Free diameter: 30 mm

Magnetic patch diameter: 8-10 mm

Magnetic composite fabrication

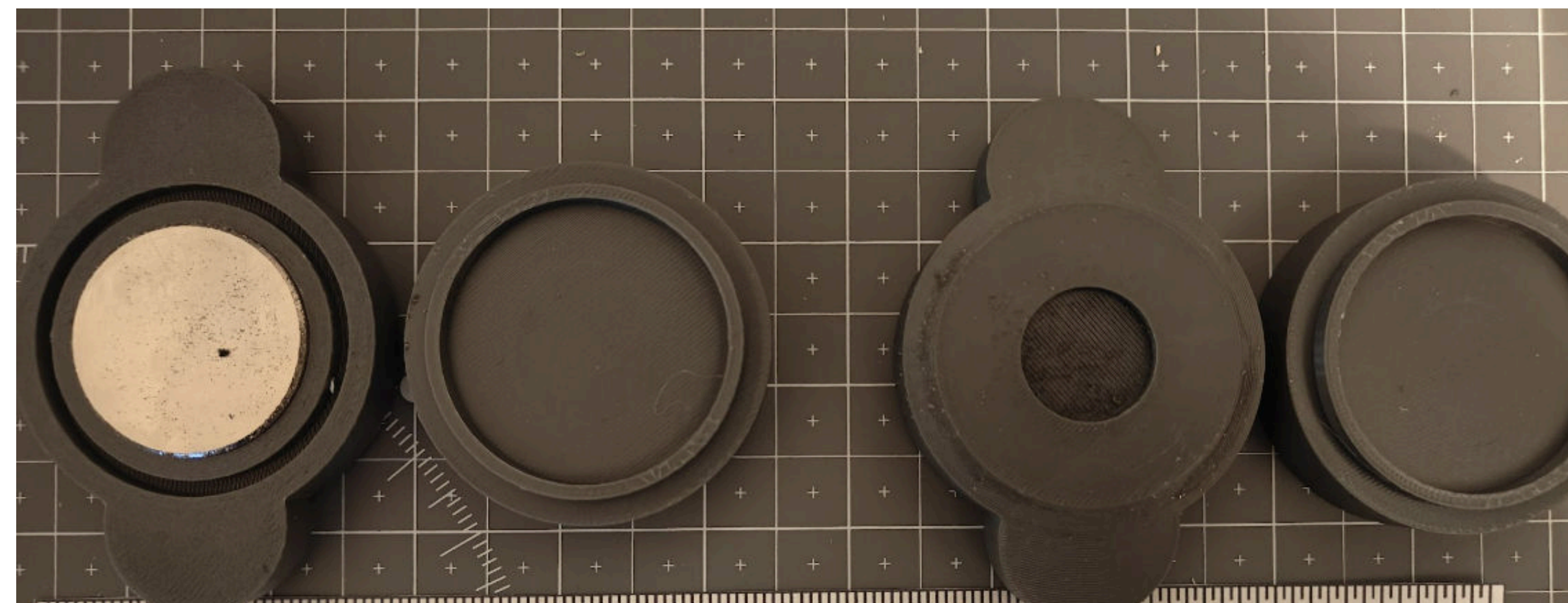


Figure 3. Molding fixture. Magnet-and-mold components.

NdFeB powder is mixed with uncured Ecoflex and cast into a mold. The filled mold is positioned between two permanent magnets, each rated for an 80 lb pull force, and cured for 24 h in the applied field to produce a permanently magnetized soft composite.

02 Conditioning and actuation

1 Condition the PCL

Temperature changes effective stiffness.

2 Apply magnetic excitation

Field and gradient act on the embedded patch.

3 Measure localized deformation

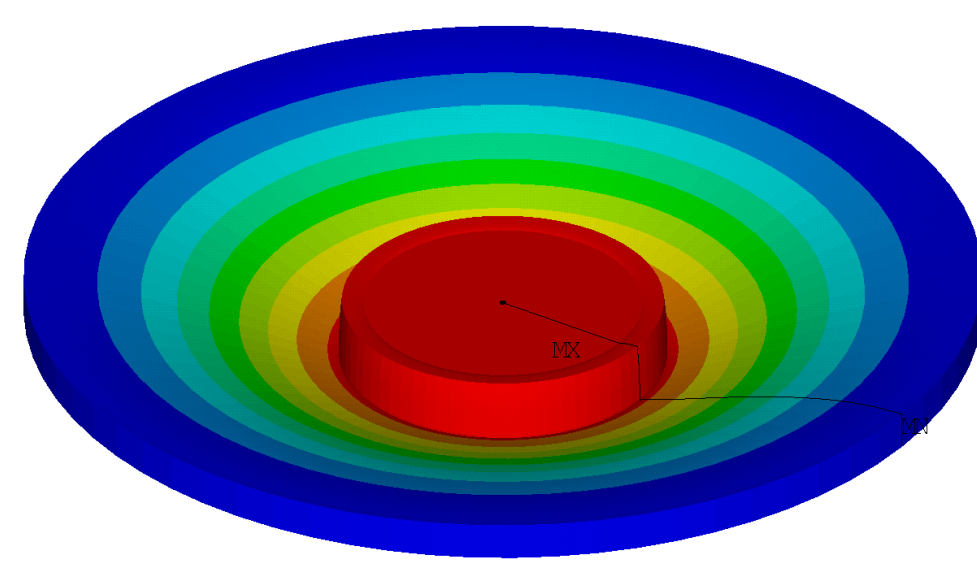
Relate magnetic input to displacement and recovery.

Working quasi-static model

$$F = k_1(T)w + k_3(T)w^3$$

F: load; T: temperature; w: displacement. Coefficients require calibration.

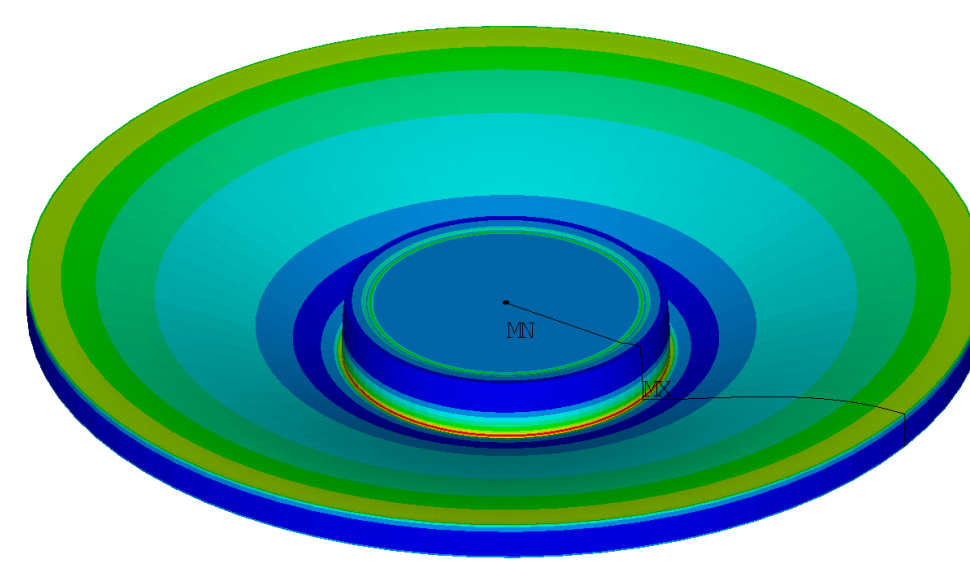
Finite-element structural analysis



0 mm 3.02 mm

Figure 4. Total displacement.

Fixed outer edge; maximum displacement magnitude: 3.02 mm.



Lower Higher

Figure 5. Stress localization.

Qualitative von Mises pattern: higher stress at the insert-membrane junction.

Model assumptions and interpretation

Axisymmetric model shown as a full disk. Outer rim fixed radially and axially; central top surface displaced downward by 3 mm. Bonded interfaces, large deflection and provisional linear-elastic properties.

Displacement is prescribed, not predicted from magnetic actuation. Stress patterns guide annular reinforcement; no quantitative stress or failure claim is made.

03 Water-bath benchmark

Initial feasibility experiment completed

Water-bath temperature typically maintained at 48-49 °C.

Experimental setup and actuation sequence

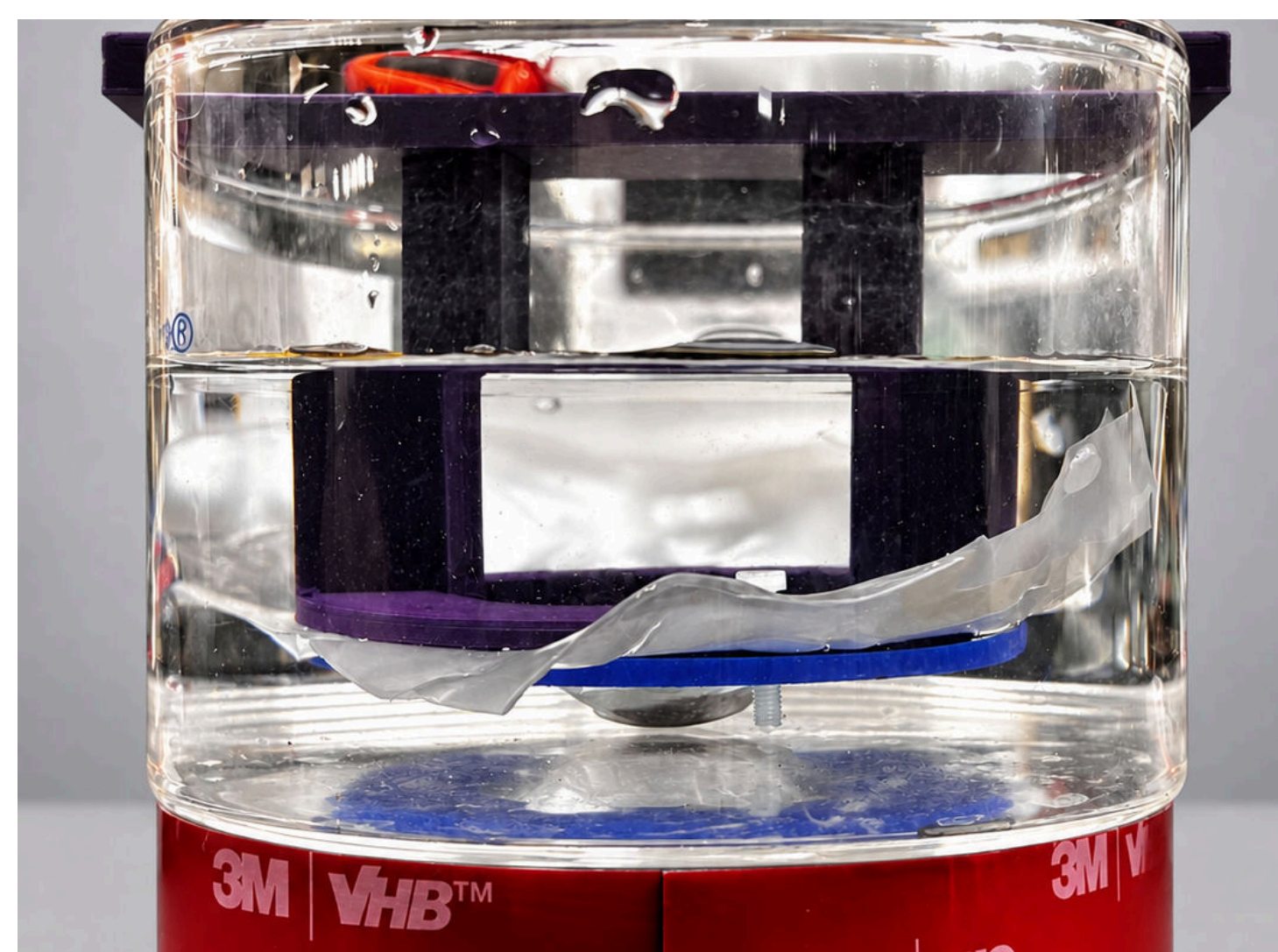


Figure 6. Electromagnetic attraction. Sample deformation in the water-bath setup under attraction.

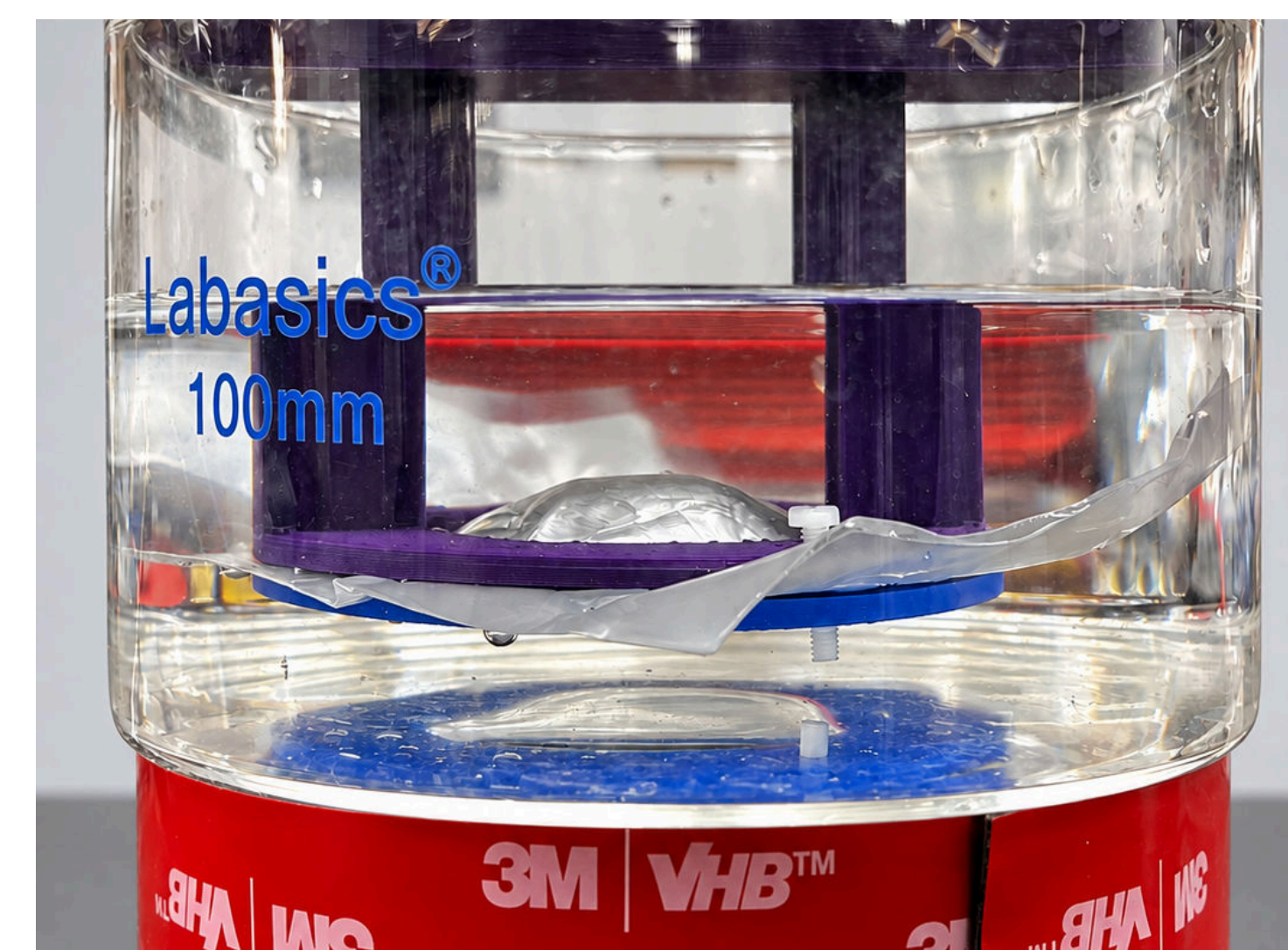


Figure 7. Electromagnetic repulsion. Sample deformation in the water-bath setup under repulsion.

Measured field-displacement response

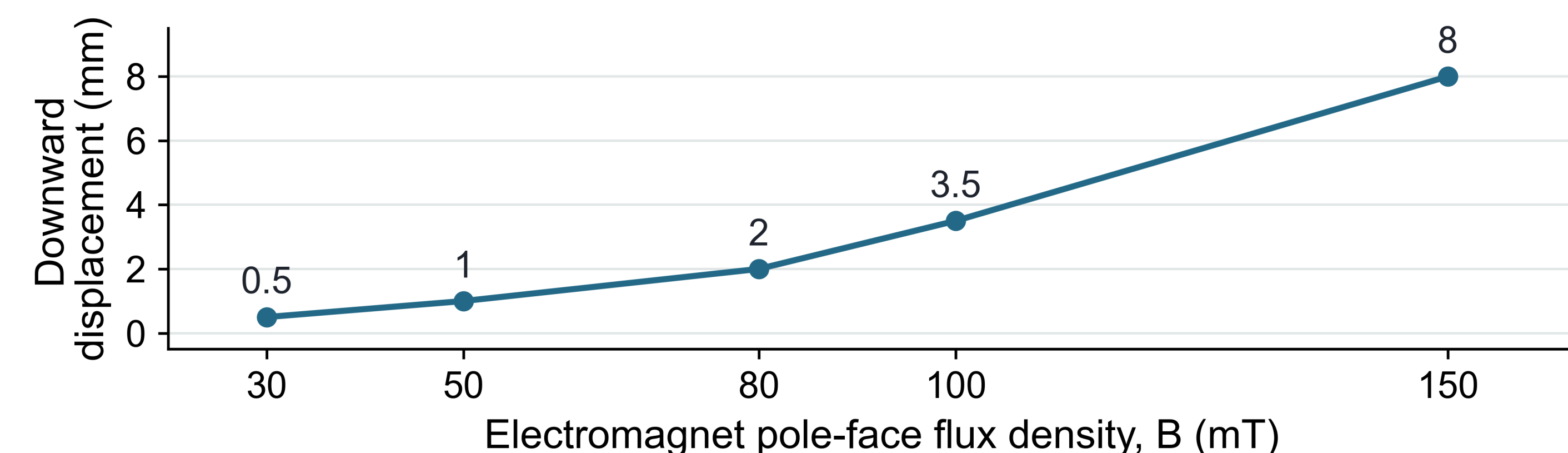


Figure 8. Measured field-displacement response.

Downward displacement under electromagnetic attraction. Markers show representative measurements, not means; connecting lines guide the eye.

Measurement protocol

B: measured at the electromagnet pole-face center, not at the sample. Center displacement is measured from images using a ruler as the scale reference, relative to the flat, unpowered state. Three repeats on one specimen per condition; representative values are plotted.

04 PDA photothermal conditioning

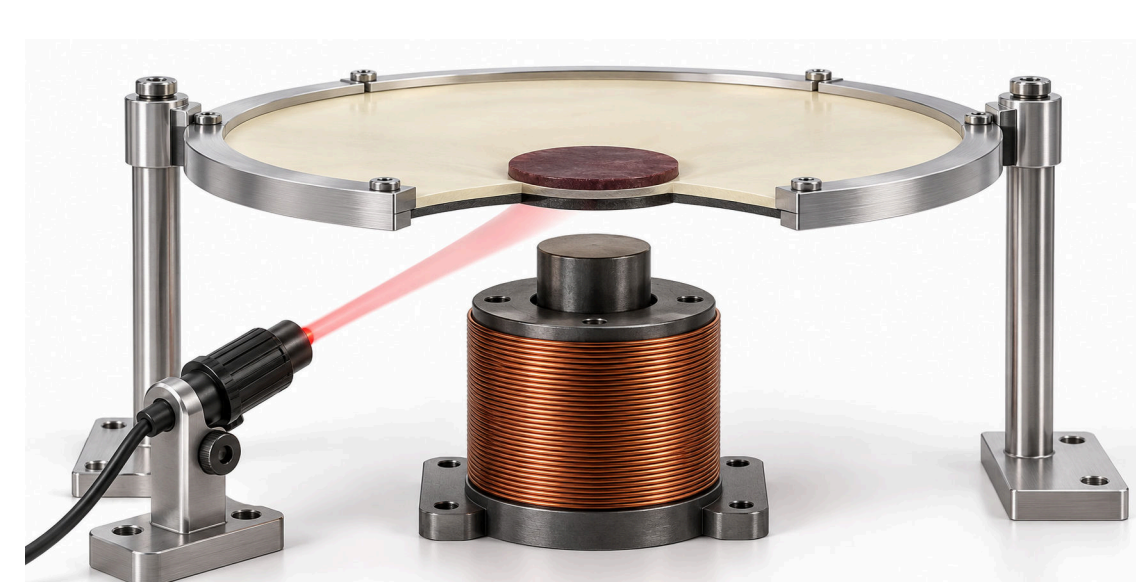


Figure 9. Proposed photothermal setup.

Concept rendering: NIR illumination from below at an oblique angle, with an electromagnet beneath the unit cell.

Planned integration study

Oblique 808 nm NIR heats the polydopamine (PDA)-coated underside. An electromagnet below the cell provides magnetic actuation.

Illustrative geometry; integrated photothermal actuation remains unvalidated.

05 Future haptic-surface interaction



Figure 10. Continuous haptic-surface concept.

Finger contact with a proposed 5 × 5 array. Array and perception tests are future work; current experimental scope is one haptic unit cell.

One continuous surface with a 5 × 5 cell array

Shape-rendering precedents include NormalTouch/TextureTouch [5] and HUGO [6]. Here, square cells would form a continuous soft sheet; local upward/downward deformation could provide distinct tactile cues.

KEY FINDINGS

At 48-49 °C, electromagnetic actuation deforms the softened PCL structure. Excessive magnetic excitation can cause cracking at the magnetic-insert/PCL interface, indicating a structural limit to the operating range.

NEXT STEPS

Integrate 808 nm PDA photothermal heating. Limit magnetic excitation and locally thicken or reinforce the annular membrane around the magnetic insert to improve load transfer and resistance to tearing.

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